Multivariable calculus · Differentiation in two variables

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

Partial derivatives, changes along a path, and a circle defined implicitly · compiled 17 October 2026

S1	Lesson: partial derivatives
S2	Lesson: total derivative
S3	Lesson: implicit curves

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1 Partial derivatives and directional slopes

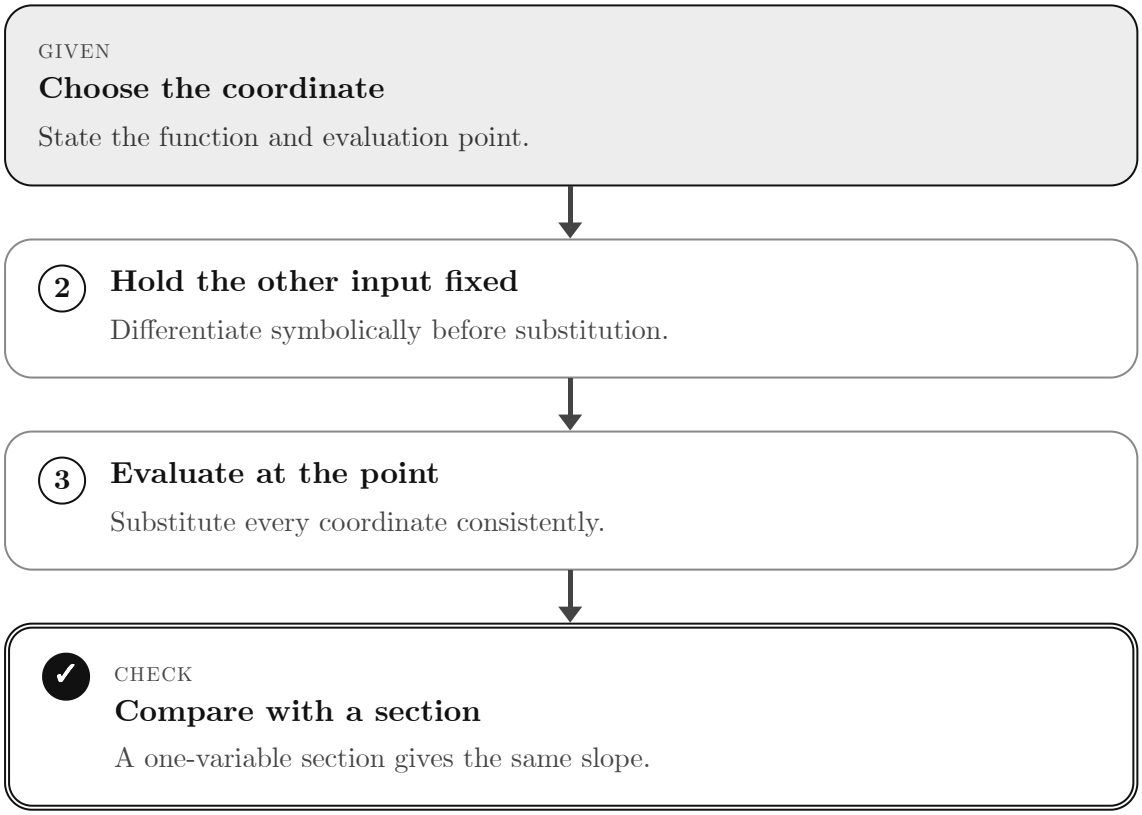


Figure F1.1. Evaluate a coordinate partial

S1

2 Total change and the chain rule

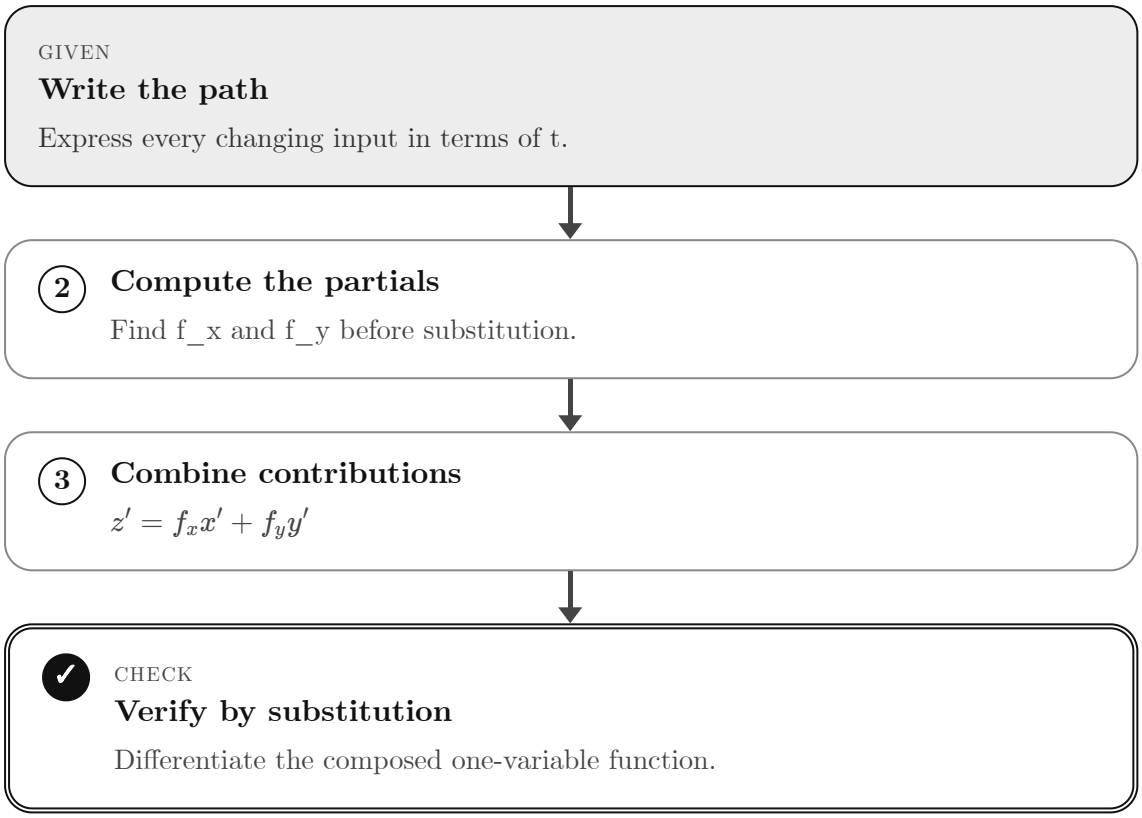


Figure F2.1. Account for both changing inputs

S2

Three related operations

Operation	What changes	What it returns
Partial derivative	One input; others fixed	Coordinate slope
Total differential	Small coordinate increments	Linear output change
Derivative along a path	All inputs specified by a parameter	Exact rate of the composition

S2

3 Implicit curves and valid tangent formulas

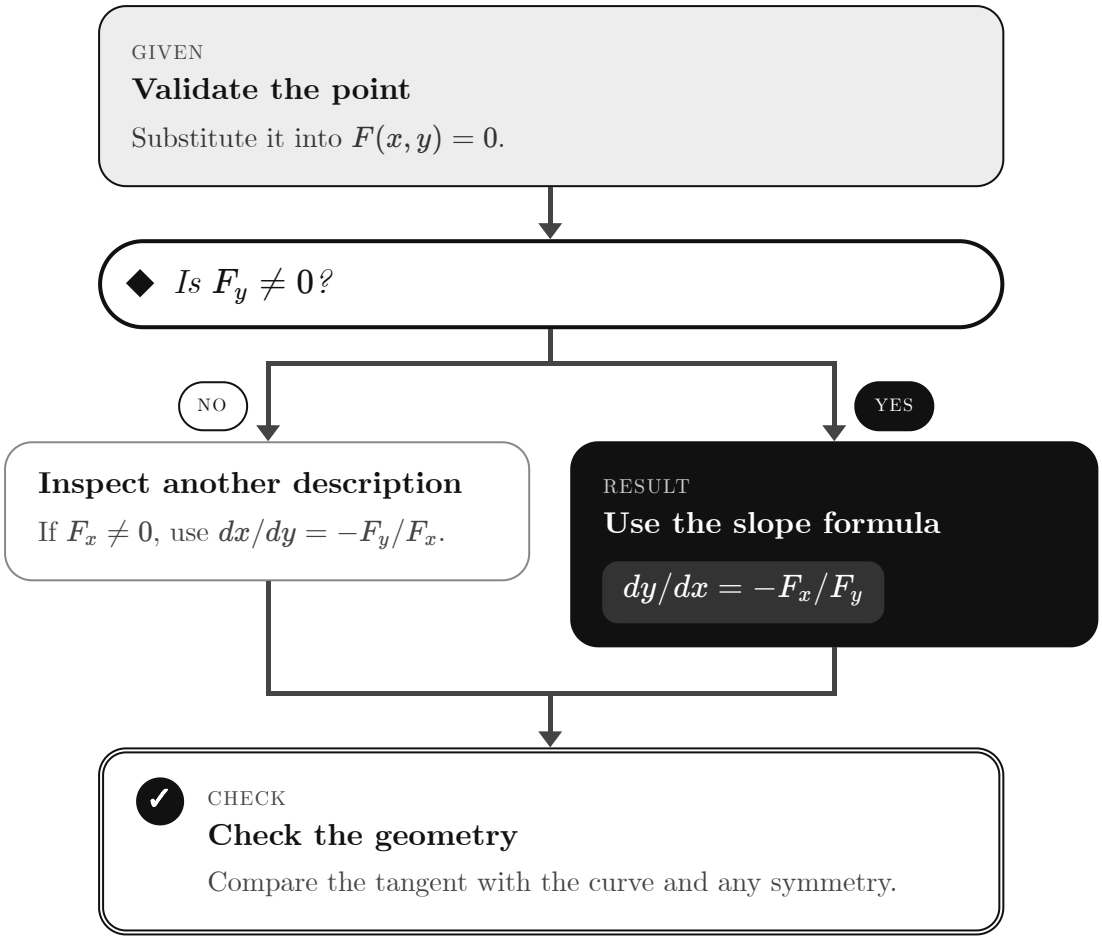


Figure F3.1. Check the implicit-derivative denominator

S3